

**ATHENS UNIVERSITY OF ECONOMICS AND BUSINESS
DEPARTMENT OF INFORMATICS**

**Wireless Networks and Mobile Communications
Fall Semester 2025-26**

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**Assignment #3 – 30 November 2025
Report submission deadline: 13 January 2026**

Note: This assignment will be implemented by groups with two students in each group.

The objective of the third assignment is to create and conduct experiments with a server and client that send parts of a large file over two wireless links. The parts are contained in separate files that are available in the following directory on MS Teams:

https://auebgr.sharepoint.com/:f:/s/msteams_fa961b/E11Z8xOsFQIMmlclMADCcmkBq_pUlgjk2UQU18sez3Llg?e=qooQTa

The files are also available in the link below:

https://auebgr-my.sharepoint.com/:f:/g/personal/vsiris_aueb_gr/EmP0EHdi0K5LvugFy5LBZtkBVaVHxUHVtyFp2ToERKpxQg?e=5K8kJ2

Devices that you will need: i) a laptop or desktop computer with Wi-Fi connectivity, ii) a smartphone, and iii) a wireless Access Point, which can be within the DSL router of your broadband connection (referred to below as “AP”) and iv) a second laptop or desktop with a wired connection to the AP. Instead of a smartphone you can use a second laptop or desktop computer and the opposite: instead of a laptop you can use a second smartphone. When conducting the measurements you are advised to keep your smartphone and laptop on a stable surface.

The report you submit must contain the topology where you conducted the measurements, the device type (smartphone or laptop/desktop), the IP addresses of the devices, the connection type (wireless or wired), and the speed of the wired (Ethernet) connections (100 Mbps or 1 Gbps) or the wireless links with the AP.

A. Implementation of server and client

Create a server that will accept requests from a client to send files. The files are named sXXX.m4s, where XXX=001 to 160, and are parts of a large file. The files are stored in the local drive of the device where the server runs. For the experiments you will need to run two servers: one server will run on the laptop (laptop with a wireless connection in scenario 1 and laptop with a wired connection in scenario 2 – the two scenarios are described below) and a server that runs on the smartphone. Hence, you will need to have two implementations of the server: one for the operating system running on the laptop and one for the operating system running on the

smartphone that you will use for the experiments. The transmission of the files will use the TCP protocol. Let us call “server A” the server running on the laptop (with a wireless connection in scenario 1 and a wired connection in scenario 2) and “server B” the server running on the smartphone.

Implement a client that requests files from the two servers. In the experiments, the client will run on the laptop (with a wired connection in scenario 1 and a wireless connection in scenario 2). The number of files that the client requests (in each “round”) from each of the two servers should be a command line option of the client program. Specifically, when the client is executed with the command «client 1 2» it should request file s001.m4s from server A, files s002.m4s & s003.m4s from server B, file s004.m4s from server A, files s005.m4s & s006.m4s from server B, etc. Hence, with the command «client 1 2», the number of files that the client obtains from server B will be double the number of files it obtains from server A.

With the command «client 1 3», the client will request file s001.m4s from server A, files s002.m4s, s003.m4s & s004.m4s from server B, file s005.m4s from server A, files s006.m4s, s007.m4s & s008.m4s from server B, etc. Hence, with the command «client 1 3», the number of files that the client obtains from server B will be triple the number of files it obtains from server A.

Similarly, with the command «client 4 1», the number of files that the client obtains from server A will be four times the number of files it obtains from server B.

The IP addresses for server A and server B can be given on the command line. Hence, the client will be executed with the command «client n_A n_B IP_A IP_B», where

- n_A: number of files that the client requests (in each round) from server A
- n_B: number of files that the client requests (in each round) from server B
- IP_A: IP address of server A
- IP_B: IP address of server B

The client must also measure the elapsed time from the point that it requests the first file up to the point that it receives (and saves in the local disk at the client) all 160 files from the two servers.

B. Experiments

Scenario 1

In scenario 1, one server will run on the laptop with a wireless connection (server A), the other server on the smartphone (server B), while the client will run on the laptop/desktop with a wired connection.

Place the laptop with server A on a stable surface near the AP. For the experiments place the smartphone where the other server - server B – runs in three different locations: one close to the AP, one far from the AP, and one in between the two locations.

For each of the three locations you selected, record the total time needed for the client to receive all 160 file parts, for the following options concerning the number of files that the client requests from server A and server B:

- client 1 5 ... (replace “...” with the IP addresses of server A and server B)

- client 1 4 ...
- client 1 3 ...
- client 1 2 ...
- client 1 1 ...
- client 2 1 ...
- client 3 1 ...
- client 4 1 ...
- client 5 1 ...

Place the results in tables and in bar chart graphs, with one table and one graph for each location where you placed the smartphone (server B).

For the laptop with the wireless connection (server A) and for each of the three locations where you placed the smartphone (server B), record the received signal strength from the AP and the TCP throughput, having as a sender the laptop with the wireless connection or smartphone that runs the server and as receiver the laptop/desktop with a wired connection that runs the client. You can measure the TCP throughput with the iperf3 tool.

Questions:

1. Based on your measurements, when is the total transfer time (time to transfer all 160 file parts) minimized?
2. Is your answer to the previous question supported/consistent with the signal strength and TCP throughput measurements?

Scenario 2

The difference between this scenario and scenario 1 is that the server (server A) runs on the laptop/desktop with a **wired connection** (in scenario 1 it was running on the laptop with a wireless connection) and the client runs on the laptop with a **wireless connection** (in scenario 1 the client was running on the laptop/desktop with a wired connection). In this scenario, as in scenario 1, server B runs on the smartphone.

Place the mobile running server B in a stable location close to the AP. For the experiments place the laptop with a wireless connection that runs the client in three locations: one close to the AP, one far from the AP, and one between the two locations.

For each of the three locations you selected, record the total time needed for the client to receive all 160 of the file parts, for the following options concerning the number of files that the client requests from server A and server B:

- client 1 5 ... (replace “...” with the IP addresses of server A and server B)
- client 1 4 ...
- client 1 3 ...
- client 1 2 ...
- client 1 1 ...

- client 2 1 ...
- client 3 1 ...
- client 4 1 ...
- client 5 1 ...

Place the results in tables and in bar chart graphs, with one table and one graph for each location where you placed the laptop with a wireless connection that runs the client.

For the smartphone (server B) and for each of the three locations where you placed the laptop with a wireless connection that runs the client, record the received signal strength from the AP and the TCP throughput, having as a sender the laptop with the wired connection that runs server A or smartphone that runs server B and as receiver the laptop with wireless connection which runs the client. You can measure the TCP throughput with the iperf3 tool.

Questions:

1. Based on your measurements, when is the total transfer time (time to transfer all 160 file parts) minimized?
2. Is your answer to the previous question supported/consistent with the signal strength and TCP throughput measurements?
3. Identify and explain any differences in your measurements in this scenario compared to the previous scenario 1.